

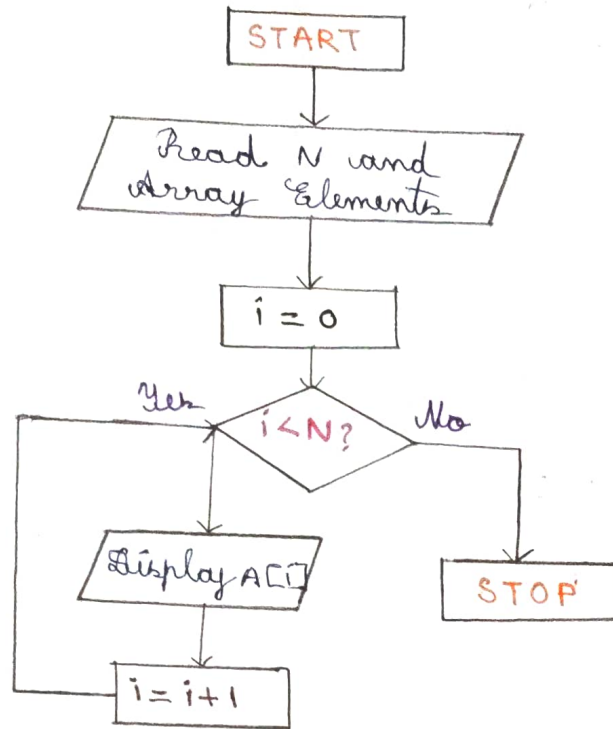
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COURSE CODE : CSA0313
COURSE NAME : DATA STRUCTURE

C05-AT3

FLOWCHART TO CODE CONVERSION

1. Traversing an Array:



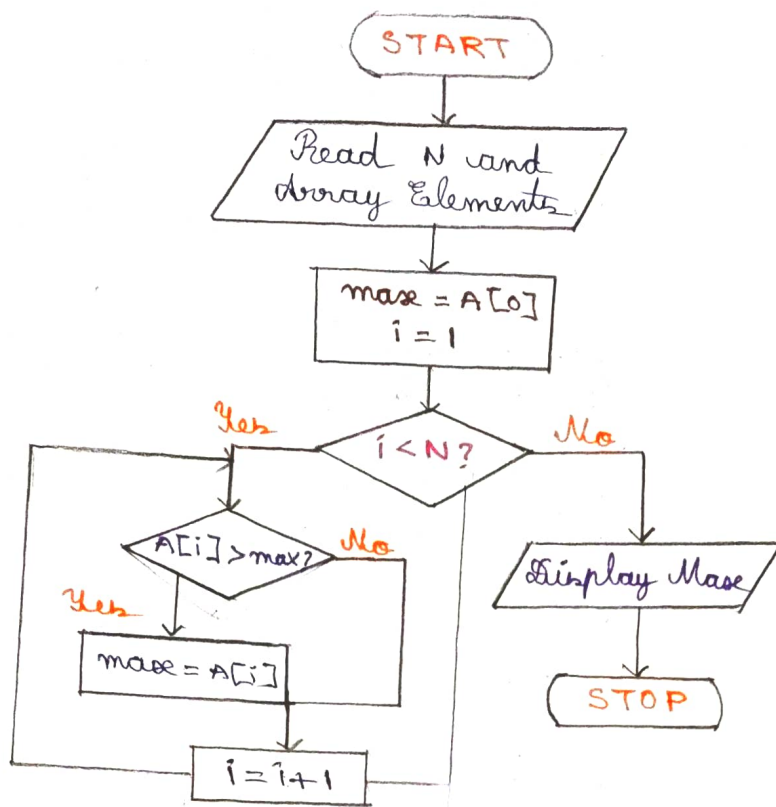
CODE

```
#include <stdio.h>
int main()
{
    int a[100], n, i;
    printf("Enter No. of Elements: ");
    scanf("%d", &n);
    printf("Enter Elements: ");
    for (i=0; i<n; i++)
        scanf("%d", &a[i]);
    printf("Array elements: ");
    for (i=0; i<n; i++)
        printf("%d", a[i]);
    return 0;
}
```

OUTPUT:

```
Enter No. of Elements: 5
Enter Elements: 10 20 30 40 50
Array Elements: 10 20 30 40 50
```

2. Largest Element:



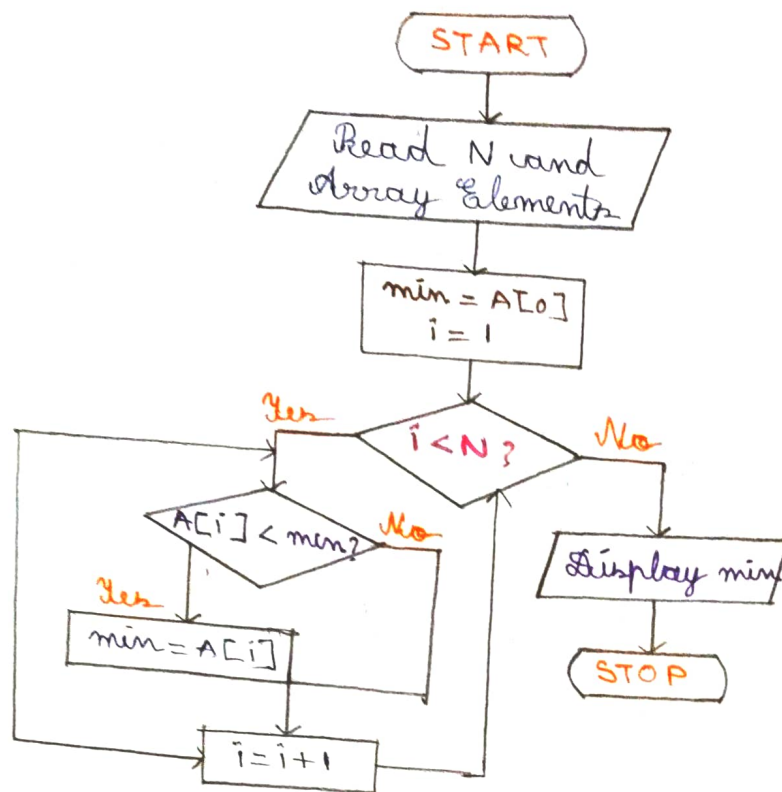
CODE:

```
#include <stdio.h>
int main()
{
    int a[100], n, i, max;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    max = a[0];
    for (i = 1; i < n; i++)
        if (a[i] > max)
            max = a[i];
    printf("Largest = %d", max);
    return 0;
}
```

OUTPUT:

```
5
10 45 23 67 38
Largest = 67
```

3. Smallest Element:



CODE:

```

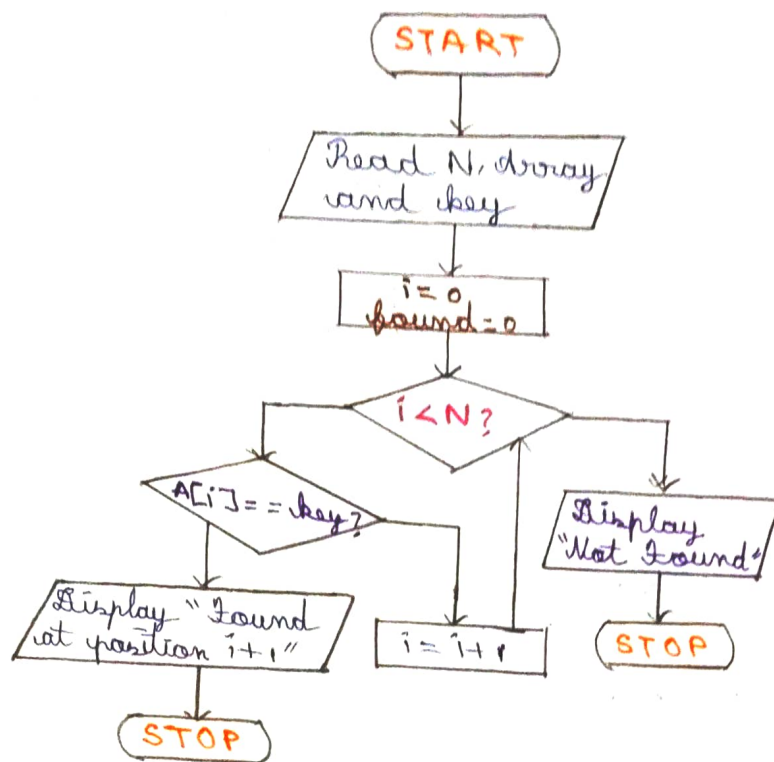
#include <stdio.h>
int main()
{
    int a[100], n, i, min;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    min = a[0];
    for (i = 1; i < n; i++)
        if (a[i] < min)
            min = a[i];
    printf("Smallest = %d", min);
    return 0;
}
  
```

OUTPUT:

```

5
27 10 15 6 38
Smallest = 6
  
```

4. Linear Search:



CODE:

```

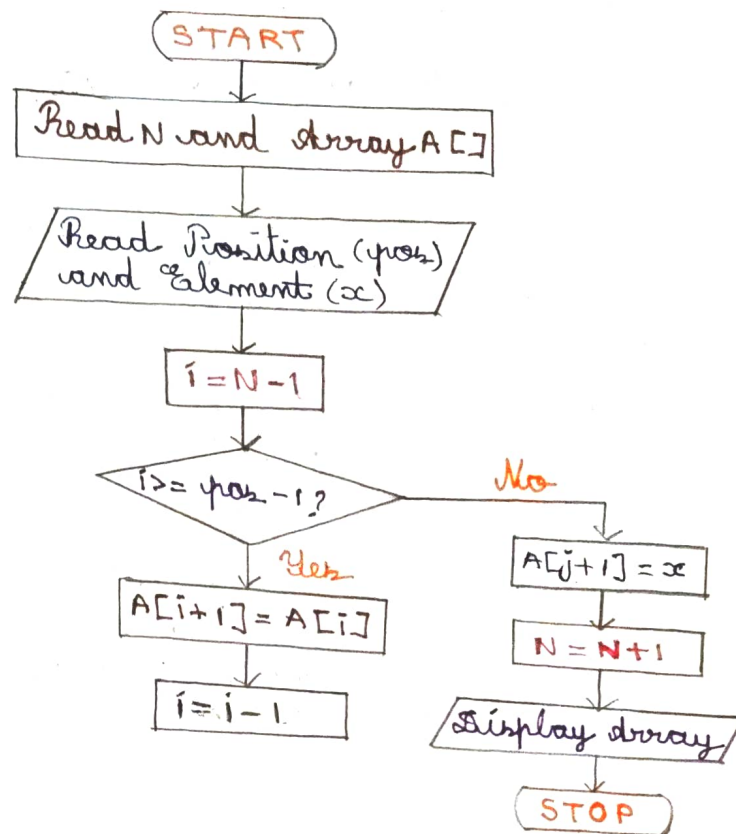
#include <stdio.h>
int main()
{
    int a[100], n, i, key;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &key);
    for (i = 0; i < n; i++)
        if (a[i] == key)
            break;
    if (i < n)
        printf("Found at %d", i+1);
    else
        printf("Not Found");
    return 0;
}
  
```

OUTPUT:

```

5
10 20 30 40 50
30
Found at 3
  
```

5. Insert Element at a Position:



CODE:

```

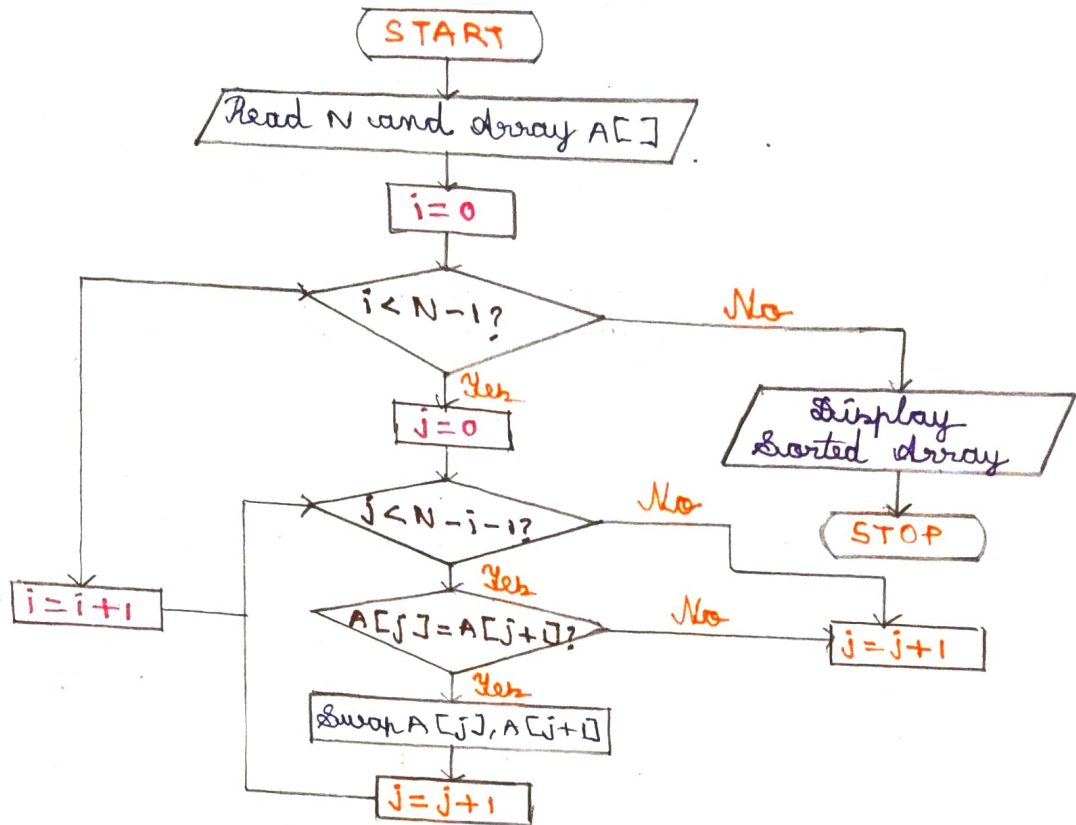
#include <stdio.h>
int main()
{
    int a[100], n, i, pos, x;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d %d", &pos, &x);
    for (i = n; i >= pos; i--)
        a[i] = a[i-1];
    a[pos] = x;
    for (i = 0; i <= n; i++)
        printf("%d ", a[i]);
    return 0;
}
  
```

OUTPUT:

```

5
10 20 30 40 50
3 25
10 20 25 30 40 50
  
```


6. Bubble Sort:



CODE:

```

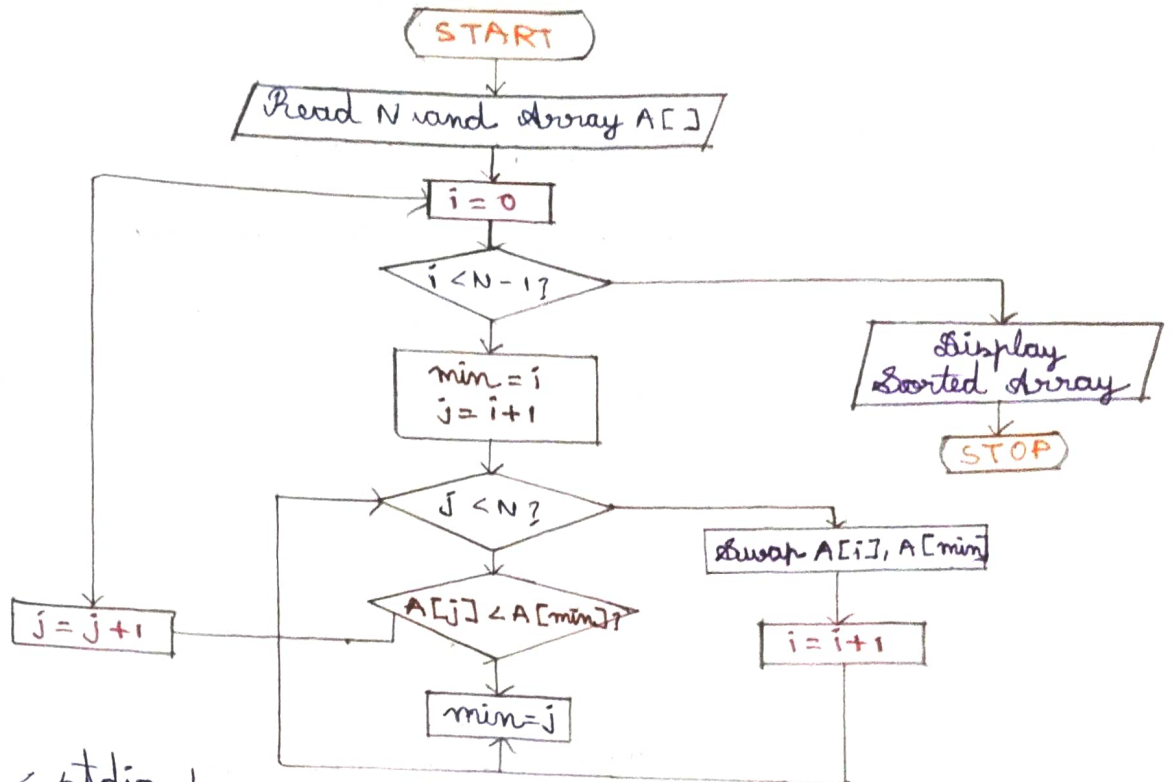
#include <stdio.h>
int main()
{
    int a[100], n, i, j, t;
    scanf("%d", &n);
    for (i=0; i<n; i++)
        scanf("%d", &a[i]);
    for (i=0; i<n-1; i++)
        for (j=0; j<n-i-1; j++)
            if (a[j] > a[j+1])
            {
                t = a[j];
                a[j] = a[j+1];
                a[j+1] = t;
            }
    for (i=0; i<n; i++)
        printf("%d ", a[i]);
    return 0;
}
  
```

OUTPUT:

```

5
50 20 40 10 30
10 20 30 40 50
  
```

7. Selection Sort:



CODE:

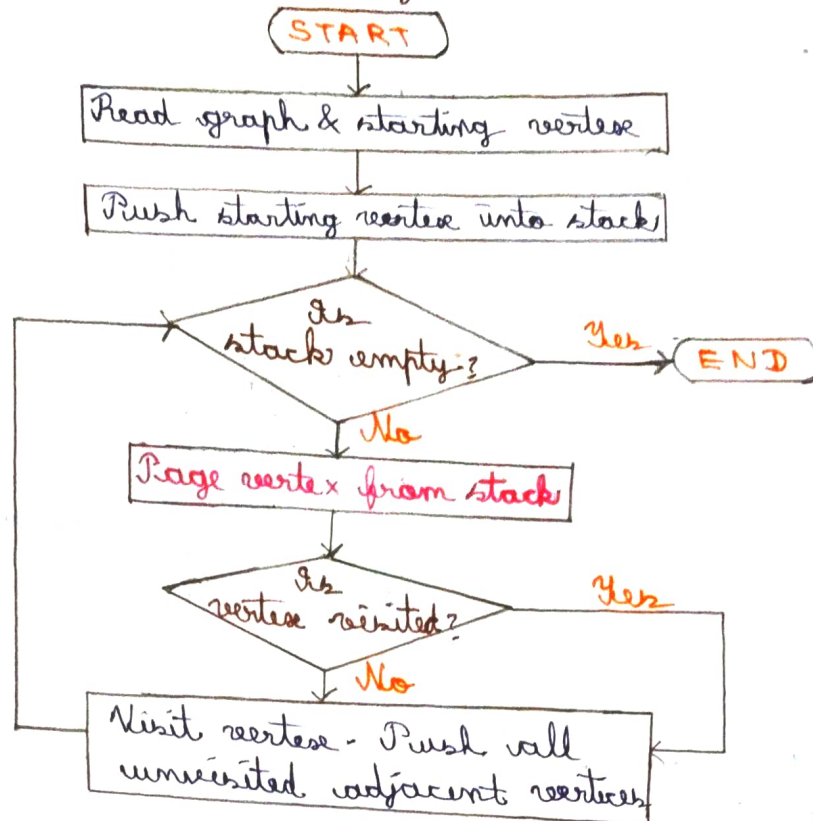
```

#include <stdio.h>
int main()
{
    int a[100], n, i, j, min, t;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for (i = 0; i < n - 1; i++)
    {
        min = i;
        for (j = i + 1; j < n; j++)
            if (a[j] < a[min])
                min = j;
        t = a[i];
        a[i] = a[min];
        a[min] = t;
    }
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
    
```

OUTPUT:

5
70 25 50 38 27
25 27 38 50 70

8. Depth - first Search (DFS) using Stack:



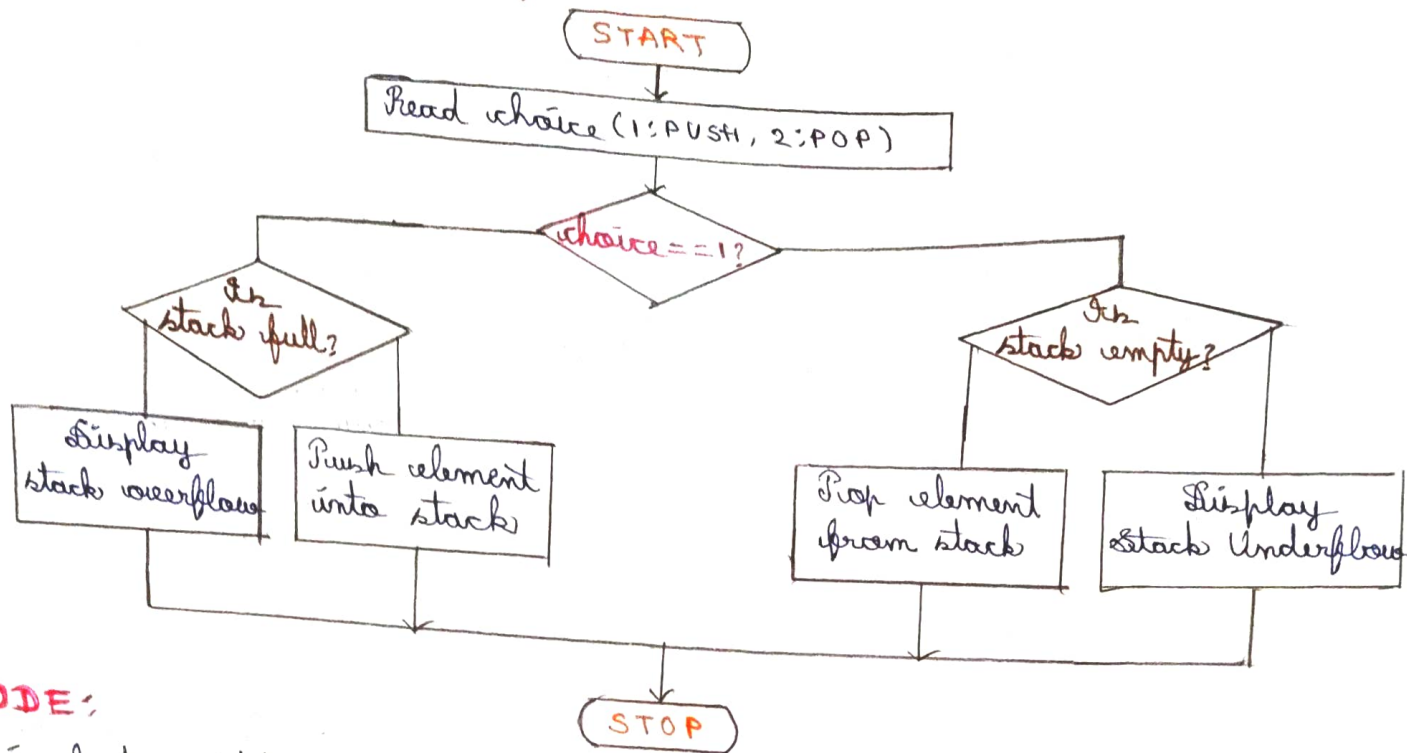
CODE:

```

#include <stdio.h>
int main()
{
    int a[10][10], v, s, st[20], top = -1, vis[10] = {0}, i, x;
    scanf("%d %d", &v, &s);
    for (i = 0; i < v; i++)
        for (x = 0; x < v; x++) scanf("%d", &a[i][x]);
    st[++top] = s;
    while (top != -1)
    {
        x = st[top--];
        if (!vis[x])
        {
            printf("%d ", x);
            vis[x] = 1;
            for (i = v - 1; i >= 0; i--)
                if (a[x][i] && !vis[i])
                    st[++top] = i;
        }
    }
}
  
```

OUTPUT:				
4	0			
0	1	1	0	
1	0	0	1	
1	0	0	1	
0	1	1	0	
0	1	3	2	

Stack Overflow & Underflow:



CODE:

```

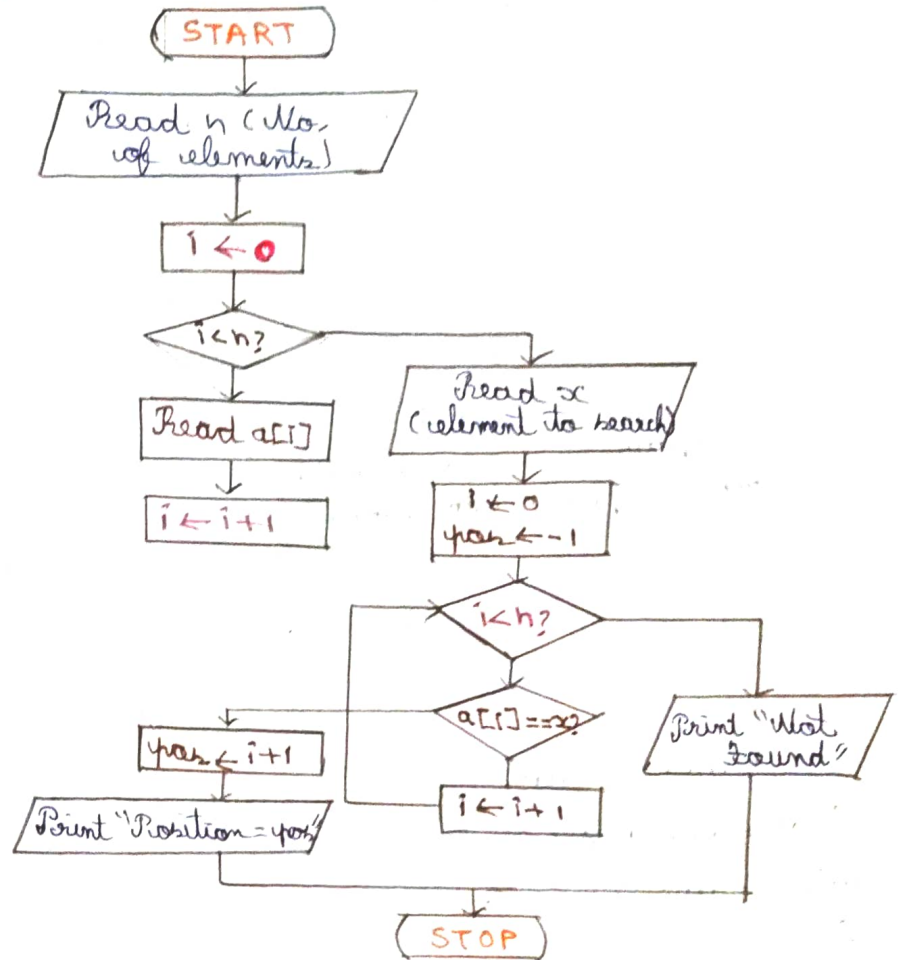
#include <stdio.h>
#define MAX 3
int main()
{
    int stack[MAX], top = -1, ch, x;
    scanf("%d", &ch);
    if (ch == 1)
    {
        scanf("%d", &x);
        if (top == MAX - 1)
            printf("Stack Overflow");
        else
        {
            stack[++top] = x;
            printf("Pushed = %d", x);
        }
    }
    else if (ch == 2)
    {
        if (top == -1)
            printf("Stack Underflow");
        else
            printf("Popped = %d", stack[top--]);
    }
    return 0;
}
  
```

OUTPUT:

```

1
10
Pushed = 10
2
Stack Underflow
  
```

10. Finding Position - Application Based:



CODE:

```

#include <stdio.h>
int main() {
    int a[100], n, x, i, pos = -1;
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for (i = 0; i < n; i++)
        if (a[i] == x)
        {
            pos = i + 1; break;
        }
    if (pos != -1)
        printf("Position = %d", pos);
    else
        printf("Not Found");
    return 0;
}
  
```

OUTPUT:

```

5
11 22 33 44 55
44
Position = 4
  
```